

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and executed system validation test plans for automotive camera systems and mechatronic ECUs, ensuring comprehensive coverage of functional and performance requirements against specifications.
- Developed and maintained test specifications aligned with system requirements, ensuring traceability throughout the project development lifecycle for embedded automotive software features.
- Contributed to the specification of test environment requirements, collaborating with test tool developers to define needs for Bench, HIL, and Vehicle testing setups.
- Implemented Python scripts to automate repetitive testing tasks, reducing manual execution time for regression test suites by approximately **30%**.
- Participated in cross-functional team meetings, including sprint planning and technical estimation sessions, to ensure timely delivery of system validation targets.
- Analyzed system logs and trace data to support troubleshooting efforts, identifying root causes for validation failures and verifying fixes.
- Containerized testing tools and services using **Docker**, deploying them to an **AWS EC2** instance for consistent and isolated test environment execution.

SKILLS

Programming Languages – Python, C++, C, C#, Java

Automotive Systems – ADAS, Mechatronic Systems, Automotive Camera Systems

Testing & Validation – System Validation, Test Specification, Test Environment, HIL, Bench Testing, Vehicle Testing, Trace Analysis, Stress Testing, Robustness Testing

Development Tools & Platforms – Jira, Confluence, Git, Jenkins, Docker, AWS (EC2, S3, CloudWatch)

Methodologies – Agile, Scrum, Systems Development Life Cycle

PROJECTS

Automotive Software Test Automation Framework

- Developed a Python-based test automation framework to execute system-level tests for automotive software features, improving test execution efficiency.
- Integrated the framework with **Jenkins** for continuous integration, enabling automated test runs on a scheduled basis and reducing manual intervention.
- Generated detailed test reports and logs, providing clear status updates on test completion and deviations to the project team.

Embedded Systems Integration Testing

- Designed and executed integration tests for mechatronic ECU systems, validating communication protocols and data flow between components.
- Performed stress and robustness testing to evaluate system performance under non-ideal conditions, identifying potential failure points.
- Collaborated with system engineers to refine test cases based on architecture design reviews and early-stage requirement analysis.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)